

# Chi-Square

## Test for Independence Attributes

**Illustration 1.** In an anti malarial campaign in a certain area, quinine was administered to 812 persons out of a total population of 3,248. The number of fever cases is shown below :

| Treatment  | Fever | No Fever | Total |
|------------|-------|----------|-------|
| Quinine    | 20    | 792      | 812   |
| No quinine | 220   | 2,216    | 2,436 |
| Total      | 240   | 3,008    | 3,248 |

Discuss the usefulness of quinine in checking malaria.

Chi-Square ( $\chi^2$ ) Test of Independence of Attributes

Q.1

| Treatment         | Fever | No Fever | (RT) Row Total |
|-------------------|-------|----------|----------------|
| Quinine           | 20    | 792      | 812            |
| Not quinine       | 220   | 2216     | 2436           |
| Column Total (CT) | 240   | 3008     | N = 3248       |

Null Hypothesis,  $H_0$ : There is no association between quinine & Malaria.  
Under  $H_0$ , Expected frequencies

$E(20) = \frac{812 \times 240}{3248} = 60$   
 $E(792) = \frac{812 \times 3008}{3248} = 752$   
 $E(220) = \frac{2436 \times 240}{3248} = 180$   
 $E(2216) = \frac{2436 \times 3008}{3248} = 2256$

Calculation of  $\chi^2$

| $f_o$               | $f_e$               | $(f_o - f_e)$ | $(f_o - f_e)^2$ | $\frac{(f_o - f_e)^2}{f_e}$                |
|---------------------|---------------------|---------------|-----------------|--------------------------------------------|
| 20                  | 60                  | -40           | 1600            | $\frac{1600}{60} = 26.6$                   |
| 792                 | 752                 | 40            | 1600            | 2.12                                       |
| 220                 | 180                 | 40            | 1600            | 8.88                                       |
| 2216                | 2256                | -40           | 1600            | 0.70                                       |
| $\Sigma f_o = 3248$ | $\Sigma f_e = 3248$ |               |                 | $\Sigma \frac{(f_o - f_e)^2}{f_e} = 38.37$ |

$\chi^2 = \sum \left[ \frac{(f_o - f_e)^2}{f_e} \right] = 38.37$

d.f. =  $\nu = (r-1)(c-1) = (2-1)(2-1) = 1$   
 LOS,  $\alpha = 0.05$

Calculated  $\chi^2_{0.05, 1} = 38.37$   
 Tabulated  $\chi^2_{0.05, 1} = 3.84$

Decision: Cal  $\chi^2 > \text{tab. } \chi^2$   
 $38.37 > 3.84$   $H_0$  reject

Two researchers adopted different sampling techniques while investigating the same group of students to find the number of students falling in different intelligence level. The results are as follows :

No. of students in each level

| Researcher | Below average | Average | Above average | Genius | Total |
|------------|---------------|---------|---------------|--------|-------|
| X          | 86            | 60      | 44            | 10     | 200   |
| Y          | 40            | 33      | 25            | 2      | 100   |
| Total      | 126           | 93      | 69            | 12     | 300   |

Would you say that the sampling techniques adopted by the two researchers are significantly different ?

### Chi-Square ( $\chi^2$ ) Test of Independence of Attributes

Q.2

| Researchers | Below Average | Average | Above Average | Genius | Total |
|-------------|---------------|---------|---------------|--------|-------|
| X           | 86            | 60      | 44            | 10     | 200   |
| Y           | 40            | 33      | 25            | 2      | 100   |
| Total       | 126           | 93      | 69            | 12     | N=300 |

Null Hypothesis,  $H_0$ : Sampling Techniques adopted by two researchers are not significantly different.

Under  $H_0$

$$E(86) = \frac{200 \times 126}{300} = 84$$

$$E(40) = \frac{100 \times 126}{300} = 42$$

$$E(60) = \frac{200 \times 93}{300} = 62$$

$$E(33) = \frac{100 \times 93}{300} = 31$$

$$E(44) = \frac{200 \times 69}{300} = 46$$

$$E(25) = \frac{100 \times 69}{300} = 23$$

$$E(10) = \frac{200 \times 12}{300} = 8$$

$$E(2) = \frac{100 \times 12}{300} = 4$$

| $f_o$              | $f_c$              | $(f_o - f_c)$ | $(f_o - f_c)^2$ | $\frac{(f_o - f_c)^2}{f_c}$                |
|--------------------|--------------------|---------------|-----------------|--------------------------------------------|
| 86                 | 84                 | 2             | 4               | 0.046                                      |
| 60                 | 62                 | -2            | 4               | 0.064                                      |
| 44                 | 46                 | -2            | 4               | 0.086                                      |
| 10                 | 08                 | 2             | 4               | 0.5                                        |
| 40                 | 42                 | -2            | 4               | 0.095                                      |
| 33                 | 31                 | 2             | 4               | 0.12                                       |
| 25                 | 23                 | 2             | 4               | 0.17                                       |
| 02                 | 04                 | -2            | 4               | 0                                          |
| $\Sigma f_o = 300$ | $\Sigma f_c = 300$ |               |                 | $\Sigma \frac{(f_o - f_c)^2}{f_c} = 0.911$ |

$$d.f. = 2 = (2-1)(4-1) = 3-1 = 2$$



## Chi-Square ( $\chi^2$ ) Test of Independence of Attributes

Q.2

| Researchers | Below Average | Average | Above Average | Genius | Total |
|-------------|---------------|---------|---------------|--------|-------|
| X           | 86            | 60      | 44            | 10     | 200   |
| Y           | 40            | 33      | 25            | 2      | 100   |
| Total       | 126           | 93      | 69            | 12     | N=300 |

Null Hypothesis,  $H_0$ : Sampling Techniques adopted by two teachers are not significantly different.

$$\chi^2 = \sum \left[ \frac{(f_o - f_e)^2}{f_e} \right] = 0.911$$

$$d.f. = 2 - 2 = 0$$

$$\text{Cal. } \chi^2_{0.05, 2} = 5.991$$

$$\text{Tab. } \chi^2_{0.05, 2} = 5.991$$

$$\text{Cal. } \chi^2 < \text{tab. } \chi^2$$

$$0.911 < 5.991$$

$H_0$  accept

| $f_o$            | $f_e$            | $(f_o - f_e)$ | $(f_o - f_e)^2$ | $\frac{(f_o - f_e)^2}{f_e}$ |
|------------------|------------------|---------------|-----------------|-----------------------------|
| 86               | 84               | 2             | 4               | 0.046                       |
| 60               | 62               | -2            | 4               | 0.064                       |
| 44               | 46               | -2            | 4               | 0.086                       |
| 10               | 08               | 2             | 4               | 0.5                         |
| 40               | 42               | -2            | 4               | 0.095                       |
| 33               | 31               | 2             | 4               | 0.012                       |
| 25               | 23               | 2             | 4               | 0                           |
| 02               | 04               | -2            | 4               | 0                           |
| $\sum f_o = 300$ | $\sum f_e = 300$ |               |                 |                             |

$$d.f. = 2 - 1 = 1$$

